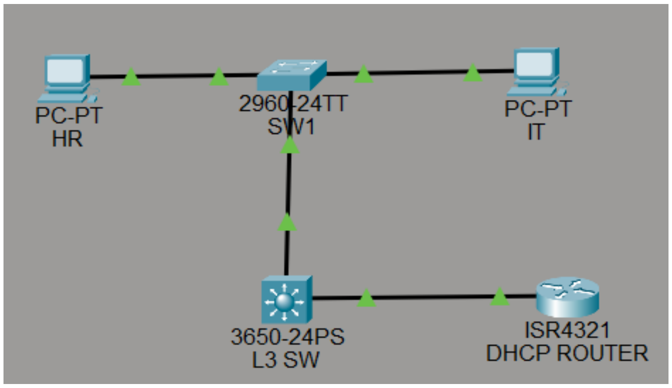
MINI PROJECT 9

Router as DHCP Server + DHCP Relay + VLANs + Static Route

DIAGRAM:



DESCRIPTION:

This project explains how a router acting as a DHCP server assigns IP addresses to hosts in multiple VLANs with static routes. It also shows how DHCP relay (ip helper-address) forwards DHCP broadcasts across different broadcast domains so each VLAN can get IP configuration.

STEP 1: Create VLANs, Access Ports & Trunk:

Create Vlan 10 and Vlan 20 on SW1 & L3 SW

Enable access port on SW for Vlan 10 and vlan 20

allow trunk on SW1 and L3 SW

STEP 2: Configure SVIs on Layer-3 Switch

```
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
interface vlan 20
ip address 192.168.20.1 255.255.255.0
 no shutdown
exit
```

```
interface g1/0/1
no switchport
ip address 192.168.50.2 255.255.255.0
exit
ip routing
```

STEP 3 : Configure Router Interface (Connected to L3 Switch):

```
interface gigabitEthernet0/0/1
ip address 192.168.50.1 255.255.255.0
no shutdown
exit
```

STEP 4: Configure IP Addressing on PCs:

PC	VLAN	IP Address	Gateway
PC-HR	10	192.168.10.10/24	192.168.10.1
PC-IT	20	192.168.20.10/24	192.168.20.1

STEP 5: Configure DHCP Pools on Router & static routes:

```
ip dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1

ip dhcp pool HR
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
exit

ip dhcp pool IT
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
exit
```

STEP 6: Configure DHCP Relay on L3 Switch:

interface vlan 10
ip helper-address 192.168.50.1
exit

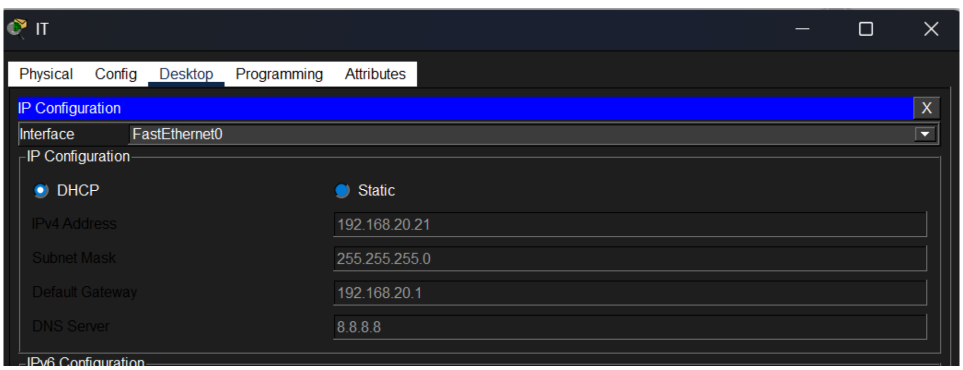
interface vlan 20
ip helper-address 192.168.50.1
exit

ip route 192.168.10.0 255.255.255.0 192.168.50.2
ip route 192.168.20.0 255.255.255.0 192.168.50.2

STEP 7 : Verification & Ping Test:

Show ip dhcp bindings

On both pcs select DHCP,



Ping test on L3 switch to HR-PC,

```
Switch#ping 192.168.10.21

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.21, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/1/3 ms
```